

MERCER COUNTY AP, WV, USAWMO: **724125**Lat: **37.298N**Lon: **81.204W**Elev: **2856**StdP: **13.24**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **03859****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.9	12.6	-4.4	4.8	15.7	-0.1	6.1	19.6	24.0	39.6	21.0	38.6	8.6	300	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	85.7	68.9	83.6	67.9	81.6	67.2	71.7	81.0	70.6	79.6	69.5	78.1	6.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.8	117.6	76.1	67.7	113.2	74.9	66.7	109.3	74.0	37.2	81.3	36.2	79.8	35.3	78.0	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.8	17.5	15.8	2.1	88.3	5.7	3.5	-2.0	90.8	-5.3	92.9	-8.5	94.8	-12.6	97.3	DB WB
			1.0	73.6	5.4	1.3	-2.9	74.5	-6.1	75.3	-9.1	76.0	-13.0	77.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.3	33.4	36.4	44.3	55.5	62.8	69.5	72.1	71.5	66.1	55.7	45.4	37.5
	DBStd	16.49	12.47	11.11	11.47	10.01	7.94	5.18	4.26	4.45	6.31	9.03	10.04	11.21
	HDD50	1915	530	398	250	61	8	0	0	0	2	51	203	412
	HDD65	4709	980	801	645	306	137	21	4	6	59	307	589	854
	CDD50	3471	15	16	73	225	404	584	687	668	484	226	65	24
	CDD65	791	0	0	3	20	68	155	226	209	92	17	1	0
	CDH74	4179	0	0	11	125	359	832	1212	1099	471	67	3	0
	CDH80	809	0	0	0	12	49	162	267	225	87	7	0	0
Wind	WSAvg	6.7	8.6	8.6	8.2	8.1	6.3	5.4	4.9	4.3	4.6	6.2	6.9	7.9
Precipitation	PrecAvg	38.6	2.6	2.7	3.4	3.5	4.1	3.5	4.3	3.2	3.2	2.6	2.6	2.9
	PrecMax	51.3	5.4	6.2	8.2	8.7	8.0	7.8	10.2	5.6	7.7	6.0	5.6	4.8
	PrecMin	30.1	0.7	0.6	1.5	0.5	1.4	0.7	1.9	0.9	0.4	0.1	0.5	0.3
	PrecStd	5.2	1.1	1.2	1.5	1.6	1.5	1.7	1.7	1.1	1.7	1.4	1.3	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.9	66.1	75.3	81.6	84.4	87.5	88.6	88.3	86.2	79.8	71.7	66.7
		MCWB	55.1	54.3	57.7	58.6	64.6	69.0	70.7	70.3	65.7	63.1	56.5	56.5
	2%	DB	59.8	61.5	69.4	77.5	80.9	84.0	85.5	84.9	82.5	75.5	67.3	62.1
		MCWB	51.4	51.7	54.5	58.0	63.7	68.3	70.1	69.2	65.8	60.7	53.9	54.6
	5%	DB	55.9	57.1	64.8	74.2	78.1	81.5	83.0	82.5	79.4	72.0	63.4	58.3
		MCWB	49.6	48.3	52.2	56.5	62.9	66.8	69.1	68.7	65.1	59.5	53.0	50.9
	10%	DB	51.7	53.0	60.9	70.5	75.1	79.0	80.6	80.2	76.2	68.5	60.3	54.3
		MCWB	45.3	45.4	50.5	55.2	61.8	66.3	68.4	68.0	64.0	58.6	51.4	48.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.5	57.2	59.2	62.8	68.2	71.8	73.7	73.2	70.3	67.4	60.2	58.9
		MCDB	62.6	63.9	71.8	75.2	79.1	82.6	83.3	83.5	79.5	72.7	67.5	64.8
	2%	WB	53.5	54.2	56.7	60.7	66.4	70.2	72.1	71.4	68.8	64.9	57.6	55.3
		MCDB	59.2	60.3	66.3	71.8	76.2	80.6	81.3	80.4	76.7	71.0	63.8	60.6
	5%	WB	50.3	50.1	54.6	59.2	65.0	68.9	70.9	70.4	67.5	62.5	55.3	52.4
		MCDB	55.1	55.0	62.8	69.6	74.3	78.6	79.8	78.9	74.6	69.4	61.0	57.3
	10%	WB	46.3	46.5	52.2	57.3	63.5	67.8	69.7	69.2	66.1	60.0	52.5	48.8
		MCDB	50.7	51.9	59.2	67.0	72.5	76.1	78.1	77.3	73.2	66.6	58.8	53.5
Mean Daily Temperature Range	5% DB	MDBR	15.1	15.8	17.0	18.5	17.0	16.4	15.3	15.4	16.2	16.2	16.4	14.3
		MCDBR	17.3	19.4	20.1	21.4	19.6	18.7	18.1	17.8	19.6	19.4	18.4	17.0
		MCWBR	12.7	12.7	11.0	9.5	8.1	7.9	7.2	6.9	7.7	8.8	10.7	12.0
	5% WB	MCDBR	15.9	16.5	17.8	17.7	17.0	16.5	15.8	15.3	15.4	15.7	15.1	16.7
		MCWBR	13.2	13.7	11.3	8.9	7.9	7.8	7.1	6.7	7.0	8.4	11.5	13.3
Clear-Sky Solar Irradiance	taub	0.287	0.295	0.318	0.352	0.391	0.415	0.443	0.452	0.354	0.330	0.299	0.282	
	taud	2.524	2.487	2.475	2.399	2.278	2.246	2.194	2.146	2.417	2.492	2.551	2.562	
	Ebn at Noon	289	300	301	294	283	275	266	261	284	283	282	284	
	Edh at Noon	24	29	32	37	42	44	46	47	34	29	24	22	
All-Sky Solar Radiation	RadAvg	662	871	1221	1576	1748	1910	1786	1659	1405	1081	800	581	
	RadStd	63	111	87	123	138	129	110	83	133	134	85	59	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65